# Few Nonlinear Columns

## Fixed-Parameter Forest Selection for Compressed Unknot Hierarchies

Research continuation prepared for
Vladimir Reshetnikov's ProveIt project

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**Abstract**

We resolve a parameterized forest-selection problem explicitly left open by the recent ProveIt research on dual-forest normalization. A candidate forest removes top handles; the remaining unsigned occurrence matrix must admit complete unit-column peeling. Columns supported on two rows give directed implications and a polynomial independent-transversal problem. A column with three unit entries introduces a two-premise activation rule.

If there are $k$ such nonlinear columns, we give an exact deterministic algorithm with cost $3^k B_{2k+1}\,\mathrm{poly}(N)$, where $N$ is the encoded input length and $B_j$ is a Bell number. The exponential factor depends on the nonlinear columns, while the number of top handles may be arbitrary. An initial event-order proof isolates at most $2k$ residual demands. A sharper reduction dispenses with event-order enumeration: one distinguished partition block records undemanded prerequisites, while other blocks share forest witnesses before matroid intersection. We also prove a weighted optimization extension and a sharper rank-layer reduction for columns satisfying the native boundary-of-boundary identity.

The accompanying exact Python implementation constructs independently checkable positive and negative certificates. Fresh source reconstruction, independent finite oracles, controlled performance comparisons, and detailed integration contracts distinguish the proved finite operator from its topological use. The result is a local algorithmic advance toward compressed hierarchies. A bound on the nonlinear parameter along every unknot hierarchy, efficient surface discovery, and global control of hierarchy visits remain separate mathematical obligations.

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# 1 Problem, contribution, and relation to the current repository

The computational target is an exact recognizer with a proved quasi-polynomial worst-case bound in the original diagram size. The present contribution addresses one definite obstruction inside that programme: *choose a legal top-handle forest so that the surviving raw attaching relations can all be eliminated*. For a fixed forest, this is already a linear-work peeling problem. Finding a good forest is the difficult part.

The newest dual-forest-normalization package proves a polynomial algorithm when every column containing a unit entry meets at most two rows. It asks explicitly for a parameterized extension to the remaining native column shape $(1, 1, 1)$ [2]. We supply that extension. The resulting theorem applies to arbitrary supplied finite occurrence systems with the stated shape; geometric authentication is required only when interpreting a successful matrix certificate as handle cancellation.

## 1.1 Exact source boundary

The inspected repository head was

`ada929648530f8689c26426e132a1b3e1c69c724`.

Its maintained `fast/` subtree has Git object identity

`697afe0edf113951dd53f8fe9b4c5011f6227b94`.

All 913 files in that subtree were checked byte-for-byte against their Git blob identities. The local baseline is the preceding commit `c6d0ec8aa0b6308310ab6c081ba29d8bc363ede5`; the inspected head adds the relative-top-handles archive and does not alter the maintained runtime. All six archives then present in `docs/incoming/` were extracted and matched to their published Git blob hashes before their contents were used. The provenance files in this package retain those identities.

The incoming work is substantial. It would be incorrect to describe its forest, stopping-set, and compression results as new results of this paper. The following table states the relevant difference in scope.

| Earlier contribution | Relation to the present result |
|---|---|
| Dual-forest normalization | Polynomial two-row-column case; general obstruction-guided search parameterized by top-forest depth. We allow nonlinear columns with fixed-parameter dependence on their number. |
| Relative top handles | Source-relative elimination and a kernel for exceptional *rows* after a fixed forest. Our parameter counts *columns*, and the forest is selected during the algorithm. |
| Positive cancellation | Exact completion by immutable column peeling, deferral, and passive-count transport. We use its completion criterion. |
| Dual forests and homological progress | Native weighted forest heuristics and an algebraic progress certificate. Our weighted theorem optimizes over *successful* forests under the nonlinear parameter bound. |
| Component-gauge cut atlases | Exact small-cut queries on supplied graph grammars; a different graph operation from selecting handle-cancellation forests. |
| Flag regularization | Compressed regular cell structures and a controlled Morse interface; it does not perform the forest search solved here. |

## 1.2 Current external mathematical baseline

Lackenby's February 2021 Davis slides announce $n^{O(\log n)}$, while his March 2021 Oxford slides state $2^{O((\log n)^3)}$ and explain the latter using a hierarchy length of order $(\log n)^2$ and

a polynomial pattern bound [4, 5]. These are two dated formulations of the announced programme and should not be silently identified.

The July 2026 preprint gives a rigorous hierarchy framework. Proposition 9.1 bounds step visits in terms of hierarchy length and pattern complexity; Theorem 14.4 supplies polynomial encoded-time cutting and parallelity-bundle information. The manuscript leaves further acceleration to future work [6]. Our local selection theorem fits between the compressed geometric construction and its subsequent cancellation test.

### 1.3 Attribution and novelty boundary

Sets generating every variable under definite-Horn closure are usually called *keys*. Symmetric all-but-one implications are studied as hypergraph Horn functions [7, 8]. Column peeling also has established connections to stopping sets and triangular matching patterns [9, 10]. Independent transversals and their algorithmic solution are classical matroid theory [11, 12].

The contribution here is the bounded-event reduction with an external forest constraint: mandatory unary source components absorb all but a bounded family of predecessor requirements; the remaining requirements can share witnesses and are handled by a set-partition reduction. A separate geometric rank invariant sharpens this reduction on authenticated native incidence data. We did not identify this exact parameterized theorem in a focused primary-source search. That search and the repository audit support the stated project contribution; they do not establish worldwide priority.

The mathematical arguments are research-draft proofs, backed by independent computational checks. They have not been formalized in a proof assistant or subjected to journal peer review. The package contains all details needed to scrutinize the claims at the finite-operator level.

## 2 Unsigned cancellation and its geometric input

### 2.1 The finite decision problem

Let $R$ be a finite set of row labels and $C$ a finite set of columns. The matrix $A = (a_{rc})$ has nonnegative integer entries and is stored sparsely. Each row either labels one edge of a finite connected undirected multigraph $G$ or is declared unavailable for top cancellation. The graph has a distinguished exterior vertex 0. Parallel edges are distinct row labels; loops cannot belong to a forest. Write $E \subseteq R$ for the eligible nonloop edges and $r = |V(G)| - 1$ for the rank.

A top choice is a spanning tree $T \subseteq E$. It removes the rows of $T$. On the remaining row set $L$, a column $c$ is a legal leaf when

$$\sum_{r \in L} a_{rc} = 1.$$

There is then a unique row $u \in L$ with $a_{uc} = 1$; delete $u$ and record the pair $(u, c)$. Continue until every row is deleted or no leaf remains. The input question is whether *some* top tree gives complete peeling. Rows unavailable for top cancellation can still be deleted by leaf pivots; excluding them from the matrix would change the problem.

**Lemma 2.1** (Saturation preserves the finite problem)**.** *Replacing every positive entry $a$ by $\min(2, a)$ preserves the legal leaf predicate on every subset of rows, and therefore preserves every peeling trace and the answer for every top tree.*

*Proof.* A sum of nonnegative integers is one exactly when one summand is one and every other summand is zero. Saturation preserves zero, one, and the property of being at least two. This applies to each column on every row subset. □

This observation does *not* license saturating counts before a parity or signed-chain calculation: an odd count of three would become even. The implementation's input digest binds the uncapped integers, and its cycle-compatibility check uses their actual parities.

**Definition 2.2** (Stopping set)**.** A nonempty subset $S \subseteq R$ is a stopping set if every column sum $\sum_{r \in S} a_{rc}$ is zero or at least two.

**Proposition 2.3** (Exact immutable obstruction)**.** *Peeling a row subset $L$ completes if and only if $L$ contains no stopping set. Its final residue is independent of the order of legal leaf choices and equals the union of all stopping subsets of $L$.*

*Proof.* A stopping set cannot lose its first row to a legal leaf: the corresponding column would already have sum at least two inside that stopping set. Consequently every stopping subset remains throughout every peeling execution. Conversely, a nonempty terminal residue has no unit column and is itself stopping. The union of stopping sets is stopping: for a column with a positive contribution to the union, one constituent stopping set already contributes at least two. The union of all stopping subsets is therefore the largest stopping residue, possibly empty. Every terminal residue contains this union, and a nonempty terminal residue is itself stopping and hence contained in it. If the union is empty, all executions complete. □

The preceding incoming reports prove that complete positive raw cancellation can always be witnessed by such immutable column peeling [1, 2]. This completion theorem is the reason a search over more complicated positive substitutions is unnecessary for the present decision problem. A certificate below uses only the original matrix entries and an ordinary tree.

## 2.2 Activation reverses the viewpoint

Call a removed row *active*. A top tree is the initial active set. A column with support $H$ can activate its row $u$ exactly when

$$a_{uc} = 1, \qquad H \setminus \{u\} \subseteq X,$$

where $X$ is the current active set. Thus it gives the Horn rule $H \setminus \{u\} \to u$ for each of its unit entries. A column with no unit entry gives no rule. A unit singleton is an unconditional rule; two-row supports give directed unary rules. Supports of three or more rows are the nonlinear part of the system.

The closure $\mathrm{cl}_A(X)$ is obtained by repeatedly applying all legal rules. Finite monotonicity shows that it exists and is independent of the application order. The tree $T$ is successful exactly when $\mathrm{cl}_A(T) = R$. This is an independent-key problem for a Horn closure, where independence is taken in an external graphic matroid.

## 2.3 Source-relative geometric interpretation

The native complementary core is a handle structure obtained from an authenticated triangulation and normal coordinate vector. After fixing the lower 0/1 forest, its surviving one-handle cocores index columns. Two-handle attaching curves give *geometric occurrence counts*; algebraic exponent sums cannot replace them. The cap graph records the two actual three-handle ends of each source two-handle, with exposed ends sent to the exterior root. Its rooted spanning trees give legal 2/3 cancellation schedules under the source-relative theorem proved in the incoming packages [2, 3].

After top cancellation, an immutable column leaf supplies a one/two cancellation in the already authenticated source model. Finite bundle boundary relations must also be included. In particular, later annular rows are not eligible cap edges and are represented as unavailable

rows. Their omission would turn an incomplete core criterion into an unsound whole-cut criterion.

The graph–matrix algorithm does not independently certify that these data arise from a manifold. Its positive output becomes a geometric certificate only after the cap graph, the occurrence matrix, and the required relative charts have been reconstructed and checked. A negative output says that this *fixed raw forest-and-peeling criterion* cannot complete. It does not say that the cut piece is not a handlebody, or that the original knot is nontrivial.

### 2.4 Why the exceptional column shape is natural

For original native one-handle cocores, a triangular source-face region has at most three actual boundary occurrences. A column containing a unit and more than two owners therefore has precisely three distinct unit owners. Moreover, the raw parity incidences obey the boundary-of-boundary identity [2]. The new source audit rechecks these properties before saturation. It also finds the same three-occurrence bound on every reconstructed finite boundary matrix in the maintained corpus. The latter is an experimental statement about those matrices, not a universal theorem about arbitrary interfaces.

The parameter $k$ will count nonlinear columns after free cancellations. The local theorem supplies no uniform polylogarithmic bound on $k$ in the original crossing number, and does not derive one from the underlying manifold type or from the number of bits per normal coordinate. Establishing useful geometric control of $k$ is one of the main next questions.

## 3 Few nonlinear columns: an exact fixed-parameter forest algorithm

Let $R$ be the row set of the immutable nonnegative occurrence matrix, and let $E \subseteq R$ label all nonloop edges of a connected undirected multigraph $G$. Parallel edges are allowed; other rows are matroid loops. A selected forest activates its rows initially. A unit-column pivot activates its last remaining row. We use the established equivalence between complete positive raw cancellation and complete unit-column peeling for this fixed source presentation.

An active column has at least one entry equal to one; a column without such an entry is inert for unit peeling. Form the ordinary implication digraph $P$ on $R$ from all active columns supported on at most two rows. A singleton unit column contributes its row to the seed set $Z$. A two-row column supported on $u, v$ contributes $v \to u$ if its entry at $u$ is one, and $u \to v$ if its entry at $v$ is one. Thus a $(1,1)$ column contributes both directions and a $(1,a)$ column with $a \geq 2$ contributes only the direction towards the unit entry.

Let $H_1, \ldots, H_k$ be the supports of the remaining active columns. Assume each consists of three distinct unit owners. Each such column contributes the three implications

$$H_i \setminus \{x\} \longrightarrow x \qquad (x \in H_i).$$

Ordinary implication closure is directed reachability in $P$, denoted $\mathrm{Reach}_P$. The full activation closure additionally uses these two-premise implications. An exceptional column can produce at most one newly activated row: after its first useful application, all three of its support rows are active.

### 3.1 Removing the closure that is free for every tree

Write $\mathrm{cl}_A(X)$ for the rows activated by unit peeling from an initial row set $X$, including all singleton-unit seeds. Compute the *free closure*

$$W = \mathrm{cl}_A(\varnothing).$$

This preprocessing involves no choice of top forest. Restrict the matrix columns to the live rows $R_0 = R \setminus W$, keeping their coefficients unchanged.

**Proposition 3.1** (Exact free-closure preprocessing)**.** *For every initially selected row set $X \subseteq R$,*

$$\mathrm{cl}_A(X) = W \cup \mathrm{cl}_{A[R_0]}(X \setminus W).$$

*Consequently the exceptional-column parameter may be measured after this preprocessing. The original cap graph and every original eligible edge must remain available for tree completion and weight optimization.*

*Proof.* The activation rule for a unit entry in row $x$ requires all other support rows of its column to have been activated. Every row of $W$ is active from every initial set $X$, by monotonicity. Once these rows are active, the rules with outputs in $R_0$ have precisely the prerequisites obtained by restricting their columns to $R_0$. Each residual activation is therefore valid in the original system after activating $W$. Conversely, remove from an original activation schedule the events whose outputs lie in $W$. Every remaining event has all of its remaining prerequisites activated earlier in the residual schedule, giving the reverse inclusion. A column used during free peeling has no live owner afterwards, so it cannot create a conflict through reuse. □

One convenient implementation retains all original row labels, marks the rows of $W$ as free seeds, and removes them only from the residual column supports. Their vertices then have no residual clauses. This keeps the original connected graph and its eligible edge set unchanged. Equivalently, discarded free rows may be retained as extension-only graph edges. A column losing one of its three unit owners becomes an ordinary two-row column, so the residual value of $k$ cannot increase. Authenticate any geometric parity identity from the uncapped source counts before capping multiplicities.

Henceforth, recompute $P, H_1, \ldots, H_k$ from the residual columns, retain the original $R, E, G$, and include every row of $W$ in $Z$. The rows of $W$ are isolated free seeds in the residual implication system. All subsequent parameters refer to this recomputed system.

**Theorem 3.2** (Fixed-parameter forest selection by exceptional columns)**.** *The existence of a spanning tree $T \subseteq E$ whose complementary rows peel completely can be decided deterministically in*

$$3^k k! B_{2k}\ \mathrm{poly}(N) = 2^{O(k \log(k+2))}\,\mathrm{poly}(N)$$

*bit operations, where $N$ is the encoded graph–matrix input length and $B_j$ is the $j$th Bell number. A successful tree and an immutable-source peeling schedule can be constructed within this bound. The algorithm enumerates no $|R|^k$ family of forest-edge choices.*

The parameter in this theorem is the number of exceptional *columns*, and the algorithm chooses the forest jointly with the peeling strategy. It is therefore distinct from an algorithm that fixes the forest first or parameterizes exceptional rows. Its proof follows from a reduction to classical independent transversals; no matroid property is assumed for the family of peelable row sets.

### 3.2 Ordered exceptional events

A *scenario* consists of a permutation $i_1, \ldots, i_k$ of the exceptional columns and an output choice $o_j \in H_{i_j}$ for each position. At event $j$, the two rows of $H_{i_j} \setminus \{o_j\}$ must already be active, and $o_j$ becomes active. An output already active is allowed; the event then has no effect. There are $3^k k!$ scenarios, including one empty scenario when $k = 0$.

**Lemma 3.3** (Complete scenario family)**.** *Every successful forest admits a complete activation schedule obeying one of these scenarios. Conversely, a legal scenario execution that activates every row certifies success of its initial forest.*

*Proof.* Take a complete activation schedule and list, in their actual order, the exceptional-column applications that activate a new row. No column appears twice in this list. Append every unused exceptional column after the original schedule finishes, choosing its output arbitrarily. These appended events are legal because all rows are already active. The converse follows directly from the implication rules: each useful exceptional event is a column with exactly one unit occurrence remaining. Ordinary directed reachability is implemented by the corresponding two-row or singleton unit columns. □

Fix a scenario and write $O = \{o_1, \dots, o_k\}$. Let $Q_1, \dots, Q_p$ be those source strongly connected components of $P$ that meet neither $Z$ nor $O$, and put

$$A_\ell = Q_\ell \cap E, \qquad U = Z \cup \bigcup_{\ell=1}^{p} Q_\ell.$$

The sets $A_\ell$ are pairwise disjoint. Every complete execution of this scenario must select a row in each $A_\ell$: no ordinary arc enters $Q_\ell$, and no chosen exceptional event outputs into it. Conversely, one selected row in $Q_\ell$ activates all of $Q_\ell$ by strong connectivity. These are obligations *conditioned on the scenario*; they need not be stopping sets of the full matrix when exceptional columns are present.

For a prerequisite $x \in H_{i_j} \setminus \{o_j\}$ at event $j$, set

$$C_j = \mathrm{Reach}_P(U \cup \{o_1, \dots, o_{j-1}\}).$$

If $x \in C_j$, no additional requirement is needed. Otherwise introduce the *predecessor demand*

$$D_{j,x} = \{e \in E : x \in \mathrm{Reach}_P(\{e\})\}. \tag{$*$}$$

There are at most $2k$ retained demands. Repeated demands may be kept as separately indexed requirements. Only earlier outputs occur in $C_j$; using future outputs would permit circular activation justifications.

**Lemma 3.4** (Separation of the residual demands)**.** *Every retained predecessor demand is disjoint from every $A_\ell$.*

*Proof.* If $e \in A_\ell \cap D_{j,x}$, then $e \in Q_\ell \subseteq U$ and there is a directed path from $e$ to $x$. Consequently $x \in C_j$, in contradiction to retention of this demand. □

**Proposition 3.5** (Exact requirements for a fixed scenario)**.** *An independent edge set $F$ supports complete activation according to the fixed scenario if and only if $F$ meets every $A_\ell$ and every retained predecessor demand.*

*Proof.* For necessity, the source-component argument gives the requirements $F \cap A_\ell \neq \varnothing$. Immediately before event $j$, every active row belongs to

$$\mathrm{Reach}_P(Z \cup F \cup \{o_1, \dots, o_{j-1}\}).$$

A prerequisite $x \notin C_j$ is not reached from $Z$ or an earlier output. Reachability from a union is the union of the individual reachability sets, so some $e \in F$ reaches $x$. This is precisely a witness in $F \cap D_{j,x}$.

For sufficiency, meeting $A_\ell$ activates every vertex of $Q_\ell$; hence all of $U$ is available. Induct on the prescribed events. A prerequisite in $C_j$ is reached from already justified sources or earlier outputs. Every other prerequisite is reached from a selected edge in its demand. Thus event $j$ is legal. After the final event every source component of $P$ contains a singleton seed, a selected edge, or a chosen output. Every component of a finite acyclic condensation is reachable from a source component, so ordinary reachability activates every row. □

### 3.3 Partitioning the few demands

The retained demands can overlap and can share a single selected edge. Requiring a different selected edge for each would be incorrect. The following elementary lemma isolates the finite combinatorics needed to handle shared witnesses.

**Lemma 3.6** (Partition reduction for an independent hitting set)**.** *Let $M$ be a finite matroid on $E$. Let $A_1, \dots, A_p$ be pairwise disjoint subsets of $E$, and let $D_1, \dots, D_q$ be subsets of $E \setminus \bigcup_\ell A_\ell$. There is an $M$-independent set meeting every $A_\ell$ and every $D_j$ if and only if, for some set partition $\pi$ of $\{1, \dots, q\}$, the family*

$$A_1, \dots, A_p, \qquad K_B = \bigcap_{j \in B} D_j \quad (B \in \pi)$$

*has an independent system of distinct representatives.*

*Proof.* Given an independent hitting set $F$, choose $e_j \in F \cap D_j$ and partition the demand indices according to equality of their chosen witnesses. For every block $B$, its common witness belongs to $K_B$; witnesses for different blocks are distinct. Also choose a representative in $F \cap A_\ell$ for each $\ell$. The latter representatives are pairwise distinct because the $A_\ell$ are disjoint, and none equals a demand witness because each $D_j$ avoids their union. All representatives form a subset of the independent set $F$. Conversely, representatives for the displayed groups meet all original demands and mandatory groups. The empty demand family uses the unique empty set partition. □

For each partition, solve the independent-transversal problem in the graphic matroid of $G$. Its groups may overlap. To reduce to the familiar graphic/partition intersection, give every membership $e \in K$ its own edge copy $(K, e)$, with the same graph endpoints as $e$. Use the graphic matroid on these copies and the partition matroid allowing at most one edge in each required group. Two copies of the same original nonloop edge are parallel and hence dependent; copies of an original loop remain loops. A common independent set of size equal to the number of groups therefore projects to an independent system of distinct original-edge representatives. There are at most $(q+1)|E|$ copies, not $|E|^q$.

Rado's independent-transversal theorem gives the exact alternative: either these groups have independent representatives or some subfamily $\mathcal{J}$ satisfies

$$\rho_G\left(\bigcup_{K \in \mathcal{J}} K\right) < |\mathcal{J}|. \tag{†}$$

Here the union is a set of original edge labels. Repeated membership does not increase rank. This is a classical matroid result; the present reduction supplies the scenario-specific groups to which it applies.

*Proof of Theorem 3.2.* Enumerate all scenarios. For each, compute its source groups and at most $2k$ retained demands. Enumerate their set partitions and solve each graphic/partition intersection problem. By Lemma 3.4, Lemma 3.6 applies; by Proposition 3.5, a successful intersection produces a successful forest. Extend it to a spanning tree. Activation is monotone in the initially selected rows, so this extension remains successful; replay peeling directly against the immutable source matrix.

Conversely, any successful tree admits a scenario by Lemma 3.3. For that scenario it meets all required groups and demands. The partition reduction therefore makes at least one of its matroid-intersection subproblems successful.

There are $3^k k!$ scenarios and at most $B_{2k}$ partitions for each. All residual constructions and matroid computations are polynomial in the encoded input size. For $k \geq 1$,

$$3^k k! B_{2k} \leq 3^k k^k (2k)^{2k} = (12k^3)^k,$$

which proves the stated bound. Scenarios and partitions can be streamed, so the producer needs only polynomial working space apart from any explicitly retained negative certificate. □

### 3.4 Independent positive and negative certificates

A positive certificate consists simply of the selected tree and the complete peeling schedule. Its checker need not trust, or reproduce, the scenario search or the matroid-intersection implementation.

For a negative certificate, record a violated inequality (†) for every scenario/partition subproblem. An independent verifier regenerates the canonical enumeration, derives each family of requirements, checks that every subproblem is covered, and verifies the reported rank defect. Graphic rank is the number of vertices minus the number of connected components of the indicated spanning subgraph, including isolated vertices. Thus verification uses only elementary graph operations and the explicit scenario rules; it does not invoke matroid-intersection optimization. The resulting certificate is fixed-parameter sized, with the same exponential parameter factor. It is not asserted to have polynomial size for unbounded $k$.

All conclusions concern the supplied immutable graph–matrix certificate class. A negative result does not prove nontriviality of the knot or exclude other handle systems, slides, subdivisions, or hierarchy certificates.

### 3.5 Optimizing the successful spanning tree

**Theorem 3.7** (Minimum-weight successful tree)**.** *Give every eligible edge a binary-encoded signed weight. An exact minimum-weight successful spanning tree, or proof that none exists, can be found with the same* $2^{O(k \log(k+2))}\,\mathrm{poly}(N)$ *parameter bound, where* $N$ *also includes the weight encoding length.*

*Proof.* Fix a scenario and a demand partition, and let $L$ be the number of required groups. Put $r = |V(G)| - 1$. If $L > r$, this subproblem is infeasible. Otherwise create its required-group copies as above and one additional dummy copy of each original edge. The first matroid is the graphic matroid on all copies. The second is a partition matroid with capacity one for each required group and capacity $r - L$ for the dummy group. Give every copy the weight of its original edge.

A common independent set of cardinality $r$ saturates every required group and the dummy capacity. Its projection is a spanning tree satisfying the group requirements, with exactly the same total weight. Conversely, a spanning tree having distinct representatives for the required groups lifts to such a common independent set: mark the representatives by their group labels and give every remaining tree edge its dummy label. Weighted matroid intersection therefore gives the exact minimum tree weight for this subproblem. Minimizing over all scenarios and partitions gives the global optimum by the completeness proof above. Signed weights are harmless because the requested cardinality is fixed. □

This weighted theorem is separate from any unweighted reference implementation. For example, negated row-occurrence counts optimize the number of occurrences removed initially among successful trees. Such a weight is a practical surrogate; it does not prove optimality for actual end-to-end recognition time.

### 3.6 Larger exceptional supports and quantitative regimes

The same proof applies to an exceptional active column with support $S_i$, size $s_i \geq 3$, and nonempty set $U_i \subseteq S_i$ of unit-coefficient owners. Its output must lie in $U_i$ and its prerequisites are all other support rows. Put

$$Q = \sum_{i=1}^{k} (s_i - 1).$$

There are $k! \prod_i |U_i|$ scenarios and at most $Q$ residual demands, giving the following extension.

**Corollary 3.8** (Parameterization by nonlinear support)**.** *Joint forest selection and complete unit peeling with arbitrary exceptional active columns is decidable in*

$$k! \left( \prod_i |U_i| \right) B_Q \operatorname{poly}(N) = 2^{O(Q \log(Q+2))} \operatorname{poly}(N).$$

*The statement remains valid for any matroid with polynomial-time independence testing in place of the graphic matroid.*

*Proof.* Each exceptional column still has at most one useful activation. Its scenario event has $s_i - 1$ prerequisites, so the preceding arguments apply verbatim. Since $|U_i| \leq s_i$ and $k \leq Q$, the displayed enumeration factor is $2^{O(Q \log(Q+2))}$. Cloning is a parallel extension of the supplied matroid, and classical matroid intersection requires only its independence oracle. □

For the triple-unit theorem, $k = O(\log N / \log\log N)$ gives polynomial time, and $k = O(\log N)$ gives $N^{O(\log\log N)}$. More generally,

$$k \log(k+2) = O((\log N)^2)$$

is sufficient for the quasipolynomial bound $N^{O(\log N)}$; for example, $k = O((\log N)^2 / \log\log N)$ suffices. These implications are conditional on the corresponding parameter bound. The finite graph–matrix theorem alone supplies no such uniform bound for all knot diagrams.

### 3.7 Authenticated cap parity preserves rank

The general algorithm above requires no compatibility between the occurrence matrix and the graph. The native cap incidence supplies an additional structure when it has been independently authenticated. Extend the graphic matroid from $E$ to the whole row set $R$ by making rows outside $E$ matroid loops. Assume that, for every column of the *full matrix used by activation*, the eligible edges with odd *uncapped* occurrence count form an even-degree subgraph of the cap graph. This includes every added finite-boundary or interface column. The distinguished exterior vertex is included in this condition, and a graph loop contributes degree two.

**Theorem 3.9** (Rank preservation of positive activation)**.** *Under this raw parity hypothesis, every legal unit-target activation adds a row in the graphic-matroid closure of its prerequisite rows. For every initial row set $X$,*

$$\mathrm{cl}_G(\mathrm{cl}_A(X)) = \mathrm{cl}_G(X), \qquad \rho_G(\mathrm{cl}_A(X)) = \rho_G(X).$$

*In particular:*

1. *every freely activated row is a graphic-matroid loop;*
2. *every successful independent seed is already a cap-graph basis;*

3. *a scenario/partition can have independent representatives only when its number $L$ of required groups equals $r = \rho_G(E)$.*

*Proof.* Suppose a column activates its unit row $e$. All its other support rows are already active. If $e$ is a matroid loop, the claim is immediate. Otherwise $e$ is a nonloop graph edge and, because its coefficient is one, belongs to the column's odd-occurrence edge set. Every nonloop edge in an even-degree finite multigraph lies on a cycle: if it were a bridge, the sum of vertex degrees on one side of that bridge would be odd. Removing $e$ from such a cycle leaves a path between its endpoints using other odd-occurrence edges of the same column. These are among the already active prerequisites. Hence $e$ belongs to their graphic span. Induction over activations proves equality of the initial and final graphic spans.

Taking $X = \varnothing$ proves the first consequence. Complete activation has rank $\rho_G(R) = \rho_G(E) = r$; if $X$ is independent it must therefore have cardinality $r$, proving the second. For the third, an independent representative system is a successful independent seed by Proposition 3.5 and Lemma 3.6. Its cardinality is exactly the number of its required groups, so $L = r$. □

The free-closure restriction in Proposition 3.1 removes only matroid loops under this authenticated hypothesis, and consequently preserves the raw even-degree identity. By contrast, replacing an odd multiplicity at least three by two can change parity. The rank argument must therefore refer to the original raw counts, although the zero/unit/multiple predicates used by the decision algorithm can still be capped at two.

**Corollary 3.10** (One Stirling layer suffices on parity-compatible inputs)**.** *For a fixed scenario with $p$ mandatory source groups and $q$ retained demands, the parity-compatible algorithm needs only partitions with*

$$|\pi| = r - p.$$

*Thus it examines $\left\{ {q \atop r-p} \right\}$ partitions, the corresponding Stirling number of the second kind, instead of all $B_q$ partitions. A scenario with $p > r$ or $p + q < r$ can be rejected immediately. A successful set of representatives is already a spanning tree when the cap graph is connected.*

*Proof.* There are $L = p + |\pi|$ required groups. The previous theorem forces $L = r$ in every feasible subproblem. Conversely, excluding the other partitions loses no solution: a successful tree generates, through the partition-reduction proof, an independent representative system; rank preservation forces that system to contain all $r$ of its tree edges. The usual conventions include one partition of the empty set into zero blocks and no partition into an invalid number of blocks. □

For a triple-unit column, the earlier source-parity audit's local classification follows from the same hypothesis [2]. Its three row elements form either a triangle of eligible edges, a parallel pair together with a matroid loop, or three matroid loops, of graphic ranks two, one, and zero respectively. Indeed, one nonloop edge is incompatible with even degrees, two nonloop edges must be parallel, and three nonloop edges must form a triangle. Only the triangle case can obtain two prerequisites directly from a forest without another activation route. This classification and rank pruning apply exactly where the raw chain identity has been verified; they are not inferred merely from an arbitrary capped matrix.

## 4 Eliminating the activation-order search

The ordered algorithm makes causality explicit, but enumerating all $k!$ orders is unnecessary. A single distinguished block can encode the prerequisites for which no initial-seed witness is requested. Forward chaining then determines whether the chosen outputs can justify the other activations. This yields the strongest bound proved in this article.

**Theorem 4.1** (Output choices and one distinguished partition block)**.** *The triple-unit forest problem is decidable, with a successful tree and peeling certificate when one exists, in*

$$3^k B_{2k+1}\,\mathrm{poly}(N) = 2^{O(k\log(k+2))}\,\mathrm{poly}(N)$$

*bit operations. For arbitrary active exceptional supports $H_i$, with unit-owner sets $U_i$ and $Q = \sum_i(|H_i| - 1)$, the bound is*

$$\left(\prod_i |U_i|\right) B_{Q+1}\,\mathrm{poly}(N).$$

*Both statements hold with an arbitrary polynomial-oracle matroid in place of the forest independence constraint.*

### 4.1 The reduced search space

Fix one unit output $o_i \in H_i$ for each exceptional column. No event order is selected. Retain only its implication $H_i \setminus \{o_i\} \to o_i$ in the nonlinear part of this branch. Write $\mathcal{C}_o$ for closure under these chosen implications and the ordinary digraph $P$.

As before, let $Q_1, \dots, Q_p$ be the source SCCs of $P$ containing neither a free seed nor any chosen output. Set

$$A_j = Q_j \cap E, \qquad U = Z \cup \bigcup_j Q_j, \qquad X = \left(\bigcup_i (H_i \setminus \{o_i\})\right) \setminus \mathrm{Reach}_P(U).$$

Repeated occurrences of the same prerequisite row now need only one index. Let $q = |X| \le Q$, and for $x \in X$ put

$$D_x = \{e \in E : x \in \mathrm{Reach}_P(\{e\})\}.$$

Every $D_x$ is disjoint from every $A_j$, by exactly the separation argument used in Lemma 3.4.

Introduce a new symbol $\star \notin R$ and enumerate set partitions $\pi$ of $X \cup \{\star\}$. Denote by $B_\star$ the unique block containing $\star$, and define

$$S_\pi = X \setminus B_\star.$$

The rows of $S_\pi$ are required to be reached from initial forest witnesses. Rows in $B_\star \setminus \{\star\}$ are *undemanded*: they are allowed to activate in any legal way, even initially. The construction never requires them to remain inactive.

For this partition, first perform the purely logical test

$$\mathcal{C}_o(U \cup S_\pi) = R. \tag{1}$$

If it fails, discard the partition. If it succeeds, seek independent distinct representatives for the groups

$$A_1, \dots, A_p, \qquad K_B = \bigcap_{x \in B} D_x \quad (B \in \pi,\ B \ne B_\star). \tag{2}$$

The same cloned graphic/partition intersection solves this subproblem. Empty intersections are immediate obstructions.

*Proof of Theorem 4.1.* For soundness, an independent representative set $F$ activates every mandatory source SCC, and therefore all of $U$. Its representative for $K_B$ reaches every row of $B$, so all of $S_\pi$ is active. The closure test then supplies a legal activation sequence for every row. Extending $F$ to a spanning tree preserves success. A literal matrix peeling replay produces the positive certificate.

For completeness, take a successful independent seed $F$ and an actual complete activation schedule. For every exceptional column used to activate a new row, choose that row as its output $o_i$. Choose any unit output for every unused column. Each chosen mandatory source SCC contains an element of $F$, since it has no incoming ordinary arc, free seed, or chosen nonlinear output.

Let $D = \mathrm{Reach}_P(Z \cup F)$ and $S = X \cap D$. We claim that $\mathcal{C}_o(U \cup S) = R$. Replay the useful exceptional events of the actual schedule. Each prerequisite of such an event either belongs to $D$ or is ordinarily reachable from an earlier useful output. In the first case it lies in $\mathrm{Reach}_P(U)$ or in $S$, by the definition of $X$; in the second it is available by induction. Every useful chosen output is consequently justified in the restricted system.

After all useful events, every prerequisite of an unused exceptional column is available as well. Indeed, completion of the original schedule places that prerequisite in $D$ or in ordinary reach from a useful output, and the same argument applies. Thus the arbitrary chosen outputs of unused columns can now be added. Every source SCC of $P$ contains a free seed, a mandatory seed, or one of these chosen outputs. All rows are therefore reached, proving the claim. This argument includes chosen outputs that were already in the initial seed; it does not presume that every selected output was a useful event in the original schedule.

For each $x \in S$, choose $f_x \in F \cap D_x$. Such a witness exists because $x \in D$ and $x \notin \mathrm{Reach}_P(U)$, with $Z \subseteq U$. Group the elements of $S$ by equality of their chosen witnesses, and put $\{\star\} \cup (X \setminus S)$ in the distinguished block. Choose a representative from $F \cap A_j$ for each mandatory group. The demand separation ensures that all these group representatives are distinct; they form an independent subset of $F$. The constructed partition passes both the closure test and the independent-transversal test.

There are $\prod_i |U_i|$ output assignments. For each, at most $Q$ distinct prerequisites remain, so there are at most $B_{Q+1}$ partitions. Everything inside one branch is polynomial in the input length. This proves both bounds and their matroid extension. □

Equivalently, first choose $S \subseteq X$, then partition its demands. The exact count

$$\sum_{s=0}^{q} \binom{q}{s} B_s = B_{q+1}$$

explains why the distinguished block absorbs both choices. In particular, there is no additional factor $2^q$ multiplying $B_q$. The two exact strategies give the combined theoretical bound

$$3^k \min\{k! B_{2k}, B_{2k+1}\} \operatorname{poly}(N).$$

This improves the explicit parameter factor but leaves its $2^{O(k \log k)}$ asymptotic class unchanged. It is not a single-exponential algorithm in $k$.

### 4.2 Rank, weights, and independent negative replay

Under the full raw parity hypothesis, a partition passing the closure test can have independent representatives only if

$$p + |\pi| - 1 = r.$$

The representative set itself is successful, so Theorem 3.9 forces it to be a basis. Therefore, for one output assignment the number of partitions that need consideration is

$$\left\{ \begin{matrix} q+1 \\ r-p+1 \end{matrix} \right\}.$$

A partition in this Stirling layer still has to pass the nonlinear closure test and the graphic independence test. The distinguished block is counted in the partition but contributes no representative and no graphic rank.

**Corollary 4.2** (A bounded deficit gives single-exponential dependence)**.** *Assume full raw parity. For each output assignment not rejected by the cardinality test, write $\delta_o = r - p_o$. If all these deficits satisfy $0 \le \delta_o \le \Delta$, then the triple-unit algorithm has running time*

$$3^k(\Delta+1)^{2k}\,\mathrm{poly}(N).$$

*For fixed $\Delta$, this is $2^{O(k)}\,\mathrm{poly}(N)$. The hypothesis concerns every non-rejected output assignment, not merely one successful assignment. If all assignments are rejected, take $\Delta = 0$.*

*Proof.* A partition of $q+1$ objects into $\delta+1$ blocks can be encoded by labeling its distinguished block zero and labeling its other blocks canonically with $1,\dots,\delta$. Giving labels to the $q$ ordinary objects gives the bound

$$\left\{ {q+1 \atop \delta+1} \right\} \le (\delta+1)^q.$$

There are at most $3^k$ output assignments and $q \le 2k$ prerequisites. The polynomial work on rejected assignments is absorbed by the same bound. The proof also gives $(\prod_i |U_i|)(\Delta+1)^Q\,\mathrm{poly}(N)$ for general supports. □

This corollary is already reflected in the implemented Stirling-layer search; it requires no additional algorithm. In particular, for polynomial-size inputs and uniformly bounded deficit, $k = O(\log n)$ gives polynomial local time, while $k = O((\log n)^2)$ gives $n^{O(\log n)}$ local time. Proving such a uniform deficit bound is a separate geometric problem.

The weighted dummy-completion construction in Theorem 3.7 applies without change to (2). Thus the stronger output-only bound also applies to exact minimum-weight successful-tree optimization. The supplied code implements the two unweighted strategies; this weighted extension is proved but not implemented.

An exact negative certificate covers every output assignment and every required distinguished partition. A failed partition has either a failed restricted-closure test, which the checker recomputes, or a Rado rank defect for (2). Canonical restricted-growth encodings identify partitions without ambiguity. A checker verifies uniqueness and the exact Bell or Stirling count, independently reconstructs each branch, and authenticates raw parity before permitting the rank restriction. Reproducing an optimizer or trusting a producer's branch log is unnecessary.

The logical closure test provides a second protection against circular justification. Chosen outputs do not become initial seeds merely because they are listed in the assignment. They must be derived from $U \cup S_\pi$ by the actual restricted rules.

# 5 Examples, obstructions, and another polynomial regime

## 5.1 A triangle is already beyond the unary theorem

Take a cap graph that is a triangle, with its three edges as rows and one column having entry one in each row. There are no free rows or ordinary implications, so $k = 1$. Every top tree has two edges; deleting those rows leaves a legal unit pivot in the third row. An algorithm restricted to two-row column supports declines this input, while the new solver handles it exactly. The input satisfies raw cap parity, and activation preserves the rank-two span of its tree.

In contrast, three parallel edges with the same one-column matrix have rank one. Any top tree removes only one row, leaving column total two; no tree succeeds. This second input does not satisfy cap parity and illustrates why the general theorem is useful independently of the native geometric hypothesis. Rank pruning must not be enabled on it.

### 5.2 Two prerequisites can have one forest witness

Let rows $d, w, x$ be three parallel edges and let $y$ be a graph loop. Use the occurrence matrix

| | $c_1$ | $c_2$ | $c_3$ | $c_4$ |
|---|---|---|---|---|
| $d$ | 1 | 1 | 2 | 0 |
| $w$ | 1 | 0 | 0 | 1 |
| $x$ | 0 | 1 | 0 | 1 |
| $y$ | 0 | 0 | 1 | 1 |

.

The ordinary rules include $d \leftrightarrow w$, $d \leftrightarrow x$, and $d \to y$. The last column is nonlinear. For its selected output $w$, the prerequisites $x$ and $y$ can both be reached from the single forest edge $d$. The set $\{d\}$ is a spanning tree and activates all rows. Demanding two distinct edges for these two prerequisites would incorrectly reject this legal branch. The partition with $x, y$ in one witness block expresses the correct requirement. This matrix is also parity-compatible: its two-row odd supports are parallel pairs, and the odd support of the last column is a parallel pair plus a loop.

### 5.3 Circular implications are not initial information

Consider four unavailable rows $a, b, c, d$, free singleton columns for $b, d$, and triple columns on $\{a, b, c\}$ and $\{a, c, d\}$. Choose $a$ and $c$ as the respective nonlinear outputs. After the free rows are activated, the only remaining rules are $c \to a$ and $a \to c$. Neither can begin. Treating both selected outputs as initially available would manufacture a false positive. Both the chronological construction and the distinguished-block closure test reject the unsupported cycle.

### 5.4 Certified matroid closure can remove the parameter bound

The preceding algorithms place a matroid constraint on an arbitrary Horn closure. A stronger, independently certified hypothesis gives another polynomial regime. Matroid Horn functions are an established subject; the following reduction is a direct application of classical matroid intersection, included as a useful design direction [13, 12].

**Proposition 5.1** (Two-matroid activation criterion)**.** *Suppose activation from seeds $F$ is exactly $\mathrm{cl}_C(Z \cup F)$ for a matroid $C$ on $R$, with polynomial-time rank testing. Let $M$ be the external seed matroid on eligible rows $E$, and put $E_0 = E \setminus Z$. There is an $M$-independent seed activating all of $R$ if and only if $M|E_0$ and $(C/Z)|E_0$ have a common independent set of size $\rho_C(R) - \rho_C(Z)$.*

*Proof.* A successful independent seed spans $C/Z$, hence contains a basis of that contraction. This subset remains independent in $M$. Conversely, a common independent set of the stated size spans $C/Z$, and adjoining $Z$ therefore activates every row. Ordinary matroid intersection decides the condition. □

One concrete certificate labels the rows by edges of an auxiliary simple chordal graph $H$, includes all its triangle implications, and checks that every additional activation rule is valid in the graphic matroid of $H$. These conditions force equality between activation and graphic closure. To see sufficiency directly, suppose the endpoints of a missing edge $e$ are joined by a path of already active edges. The path and $e$ form a cycle. For a triangle, its rule activates $e$. A longer cycle has a chord: first derive that chord along the smaller cycle not containing $e$, then derive $e$ along the other smaller cycle. Induction proves the claim. Every allowed rule preserves graphic span, giving the reverse containment.

This class permits arbitrarily many nonlinear columns. For example, complete auxiliary graphs contain quadratically many edges and cubically many triangle columns, while the certified closure test is polynomial. The sufficient certificate is stronger than checking that each individual column has even cap parity.

**Example 5.2** (Parity does not imply the matroid exchange axiom)**.** Use the six edges of $K_4$ as rows, but supply only the three triangle columns $124, 134, 234$, omitting 123. Every column is parity-compatible. Let $X = \{12, 13\}$, $a = 14$, and $b = 23$. Then

$$\mathrm{cl}_A(X) = X, \qquad b \in \mathrm{cl}_A(X \cup \{a\}), \qquad a \notin \mathrm{cl}_A(X \cup \{b\}).$$

Indeed, adding 14 derives 24, then 34, then 23. Adding 23 leaves just the three edges of the omitted triangle, from which no supplied rule fires. This violates the exchange axiom for matroid closure.

The example separates a useful proved rank invariant from the much stronger closure hypothesis in Proposition 5.1. Sampling exchange identities, or verifying one parity identity per column, cannot replace a complete certificate for that hypothesis.

### 5.5 A family with fixed nonlinear parameter and unbounded top rank

Begin with the one-column triangle and attach $h$ new vertices to its root, each by two parallel edges. Give each parallel pair its own two-unit column, with no additional incidences. The connected graph has rank $h+2$ and exactly $3 \cdot 2^h$ spanning trees, while $k = 1$ for every $h$. Every spanning tree selects two triangle edges and one edge of each pair; all complementary rows peel immediately. The fixed-parameter theorem treats the entire family in polynomial encoded time. This is a structural separation from enumerating all top trees, not an experimental claim that the existing obstruction-guided search explores all $3 \cdot 2^h$ trees. Its first candidate can already succeed.

## 6 Exact implementation and a conservative integration contract

### 6.1 Two producers, separate replay

The package includes a standard-library Python implementation of both the ordered-event and distinguished-block algorithms. The first exposes `solve`; the second exposes `solve_output`. Their independent checkers expose `verify_result` and `verify_output`. They consume the same strict sparse schema:

```
problem = {
    "vertices": 3,
    "columns": 1,
    "edges": [[0, 1], [1, 2], [2, 0]],
    "rows": [[[0, 1]], [[0, 1]], [[0, 1]]],
}
```

Row labels are list positions. Each edge is either its two endpoint labels or `None`. Counts are positive integers, columns within one row are distinct, and column labels lie in the declared namespace. Booleans are rejected where integer labels or counts are required. Counts are bound to the result by a canonical SHA-256 digest using their original hexadecimal integer encodings. This avoids mistaking two inputs with identical saturated matrices but different raw parity for the same authenticated source.

The producer compiles free closure, ordinary reachability, and the nonlinear supports. Its graphic/partition intersection uses shortest augmenting paths on cloned edge memberships. This classical primitive is adapted from the inspected dual-forest-normalization package and is credited in the source. The new reductions supply overlapping demand groups and their independent certificate coverage.

Positive replay checks the full spanning tree and literal column leaves against the supplied matrix. Negative replay independently reconstructs the logical branch data, uses elementary graphic rank, and validates complete enumeration coverage. The checker modules import no producer, SCC optimizer, or matroid-intersection optimizer. Tests deliberately disable the producer primitives during replay, as well as corrupting tree entries, pivots, ranks, output assignments, and partition coverage.

### 6.2 Result meanings and finite budgets

| Result | Exact meaning |
|---|---|
| `SUCCESS` | A verified spanning tree and full immutable peeling trace for this input. |
| `NO_FOREST` | This input has no successful top tree under the fixed raw peeling criterion. The attached complete obstruction is checked. |
| `INCONCLUSIVE` | A stated resource allowance or recursion limit interrupted discovery; no complete mathematical result is asserted. |

The implementation can prove some negatives before exceptional search. A disconnected eligible graph has no spanning tree. If activation even from *all* eligible edges leaves a nonempty residue, monotonicity rules out every tree; that immutable stopping residue is a short obstruction. These tests are exact and retain their own replay rules.

Structural-work, branch, and partition counters make research caps explicit. They are not operating-system wall-time or memory limits: input parsing, hashing, allocations, and big-integer operations have their ordinary implementation costs. An interrupted search returns `INCONCLUSIVE`, not a partial negative certificate. The theoretical algorithms can stream their searches; the supplied producers retain explicit negative branches, so actual memory can scale with the fixed-parameter certificate size.

### 6.3 Where to place this operator in the recognizer

The appropriate insertion point follows authenticated normal-cut geometry and a chosen lower forest. Preserve the existing inexpensive top choices, including maximum raw-occurrence weight, because those choices often succeed without exceptional search. If their immutable peeling tests fail, the new operator can be used as an exact bounded fallback on the reconstructed full matrix. The polynomial two-row case remains available. Measured results in Section 7 determine practical dispatch priorities; the asymptotic theorem does not make every Python branch faster.

### 6.4 A polynomial incumbent prefix retains the FPT guarantee

The optional `integration/bounded_prefix.py` supplies one concrete dispatcher: run the existing stopping-set search for at most $|R| + 1$ visited search nodes, emit a complete result for independent replay if it finishes, and otherwise invoke `solve_output`. The companion dispatcher checker binds the input digest and calls the appropriate independent certificate verifier. It does not trust an earlier producer merely because the earlier implementation is the incumbent.

Each incumbent node performs a polynomial amount of graph and peeling work, so this prefix has a polynomial total bound. If it completes, its answer is exact. Otherwise the remaining cost is at most the bound in Theorem 4.1. The resulting abstract complete dispatcher therefore retains $3^k B_{2k+1}\,\mathrm{poly}(N)$ complexity while allowing easy cases to finish in the incumbent. A requested cap on the fallback may still produce `INCONCLUSIVE`; the full theoretical decision bound refers to the abstract unrestricted fallback. The Python reference can still reach its recursion limit and return `INCONCLUSIVE` even when no explicit work allowance is supplied.

The particular linear node allowance is a reproducible conservative choice, not a proved optimum. An obviously oversized disconnected vertex namespace routes directly to the new sparse certificate producer. This dispatcher is delivered as an integration proposal and is tested separately from the two new standalone algorithms.

The integration adapter must preserve five pieces of evidence:

1. the triangulation and normal-coordinate data from which the cut was reconstructed, including the selected lower forest;
2. the cap-edge map and exterior root, with loops and unavailable rows represented distinctly;
3. uncapped geometric occurrence counts, including every finite boundary relation used by the requested completion criterion;
4. the map from dense row labels back to native handles and annular attachments, and the source-relative charts that legalize cancellations;
5. the original marked boundary information needed by subsequent hierarchy steps.

After a verified positive matrix result, translate the tree to a rooted 2/3 schedule and the peeling trace to 1/2 cancellations, then replay them through the existing geometric certificate checker. A finite-matrix negative should only disable that particular source-relative route. It must not become a negative unknot verdict. If a cap is reached, continue the enclosing certified search according to its existing semantics.

The archive deliberately supplies a research module and a precise adapter contract. It does not silently replace maintained recognizer defaults, assert that an annular relation is an actual top cap, or claim to have implemented the missing general marked-homeomorphism compiler.

### 6.5 Verification boundary and code limitations

The independent verifier still shares the input parser and canonical digest implementation, which form a small explicit trusted base. It checks the finite mathematical predicate and assumes the caller supplied the authenticated graph–matrix record. It is not a secure parser for arbitrary untrusted files under adversarial memory limits, and no such claim is needed for the present research interface.

The exact weighted theorem and the certified matroid-closure fast path are proved mathematical extensions. The deliverable implements the two unweighted forest searches, their complete certificate checks, source reconstruction studies, and reproducible finite experiments. It contains no Lean formalization, new theorem about global normal-surface discovery, or complete quasi-polynomial unknot recognizer.

## 7 Source reconstruction, validation, and measured performance

The experiments answer three separate questions: whether the finite algorithms agree with independent exhaustive oracles; whether the native source compiler supplies the claimed

graph–matrix inputs; and how the exact strategies compare on the same saved instances. None of these experiments proves a worst-case parameter bound for knot diagrams.

### 7.1 Finite oracle and certificate validation

The ordered suite contains 16 test methods, 5,129 forest-oracle comparisons, and 800 independently brute-forced cloned-Rado comparisons. The output-assignment suite contains nine methods and 6,024 forest-oracle comparisons. Both finish with zero failures, errors, or skips. These are algorithm–instance comparisons: the suites intentionally overlap, so their sum is not a count of distinct mathematical inputs.

The exhaustive part ranges over all $3^{3\cdot 2} = 729$ capped three-row, two-column matrices on four different cap geometries, for 2,916 comparisons per included sweep. Independent oracles enumerate top trees and, in the designated tests, all legal positive raw pivots. Random tests cover unit triples, larger exceptional supports, nonunit entries, loops, and unavailable rows. Focused controls cover shared prerequisite witnesses, initially free graph edges, unsupported cyclic dependencies, complete-negative coverage, and the distinguished block's undemanded semantics.

The raw-parity tests deliberately use counts three and two. They verify that saturation cannot authorize a rank shortcut and that changing an uncapped integer invalidates a source-bound certificate even when the unit-peeling predicate is unchanged. Additional checks exercise a 16,384-bit multiplicity and a comparably large disconnected vertex namespace without expanding that namespace.

### 7.2 A fresh native reconstruction

The maintained marked-parallelity corpus contains 106 cases. We add three interior-loop cuts, five stellar-subdivision cases, and three nested interior-subdivision cases from the inspected incoming geometric fixtures. This gives 117 triangulation/normal-vector cases, each producing a source matrix and a full finite-boundary matrix: 234 correlated finite questions.

For each case the experiment reconstructs the complementary handle core, fixes and records the lower forest, counts the original attaching occurrences, and independently derives cap endpoints from the source triangulation representatives. The endpoint map is compared with the incoming cap compiler. Annular crossings and finite boundary relations are also reconstructed independently and compared with the earlier boundary adapter after saturation. The saved record includes actual face pairings, normal coordinates, row maps, raw profiles, source hashes, and exact reference decisions.

Every one of the 234 full raw inputs satisfies the cap-parity condition. All maintained full-boundary columns have raw total at most three, and every active exceptional column is a unit triple. Fifteen maintained full-boundary cases contain sixteen single-owner columns with raw count three. Their owners must be matroid loops because these particular raw inputs satisfy parity, so saturation does not alter cap parity for these single-owner columns. They demonstrate raw-source fidelity and digest binding; the mixed-column controls above demonstrate parity sensitivity in general. Only the uncapped reconstruction is distributed as the authoritative input bank.

The independent all-top-tree oracle completes on every maintained source and full-boundary matrix. It also completes on every supplementary case except the largest stellar input, where its explicit 250,000-subset cap is recorded. The older exact stopping-set search completes on all 117 cases, including that largest input, and provides a checked reference answer there. A capped exhaustive oracle is never reported as complete.

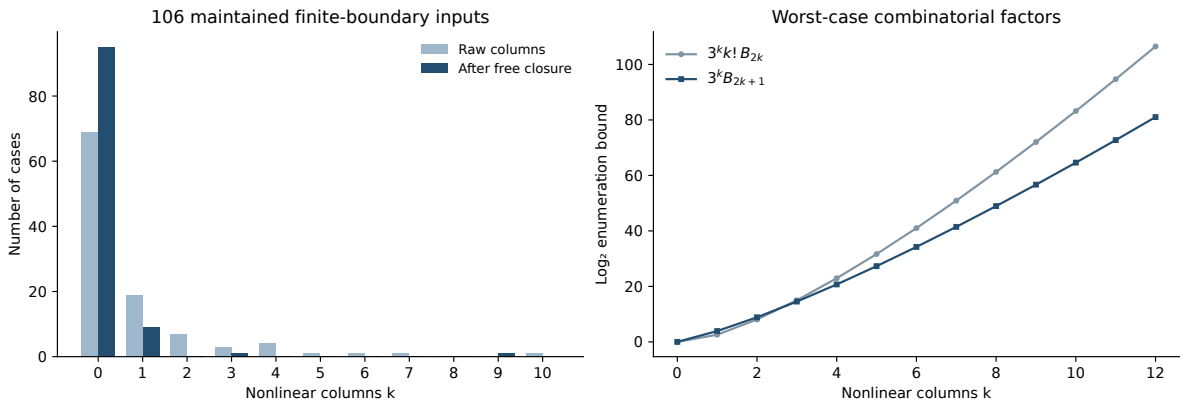



Figure 1: Left: measured raw and residual nonlinear parameters on the 106 maintained full-boundary inputs. Right: exact general enumeration factors before data-dependent rank and closure pruning. The right panel depicts combinatorial upper bounds, not measured running times.

| Strategy | Maintained matrix | Success | No forest | Inconclusive |
|---|---|---|---|---|
| Earlier two-row solver | Source | 81 | 0 | 25 |
| Earlier two-row solver | Full boundary | 67 | 2 | 37 |
| Earlier stopping search | Source | 100 | 6 | 0 |
| Earlier stopping search | Full boundary | 92 | 14 | 0 |
| Either new exact strategy | Source | 100 | 6 | 0 |
| Either new exact strategy | Full boundary | 92 | 14 | 0 |

Thus the new methods resolve the nonlinear cases declined by the old polynomial special-case solver. They reproduce the coverage already obtained by the previous general exact criterion. This experiment does not add new whole-knot recognitions or increase the maintained full-cut success count beyond 92.

Free closure substantially changes the parameter distribution. In the maintained source matrices the residual $k$ distribution is $0:98,\ 1:7,\ 2:1$. In the full boundary matrices it is $0:95,\ 1:9,\ 3:1,\ 9:1$. The large-parameter negative is rejected by a short immutable obstruction before exceptional enumeration. These observations justify the preprocessing, but are not evidence that every future hierarchy has logarithmically many nonlinear columns.

### 7.3 Supplementary native stress cases and resource ceilings

At a one-million structural-work allowance, the ordered strategy completes 228 of 234 questions: source 107/7/3 and full boundary 99/15/3, where the entries denote success, no forest, and inconclusive. The output-only strategy completes 232: source 109/7/1 and full boundary 101/15/1. Every completed result is independently replayed and agrees with the earlier exact reference. The only remaining output-only cap is the largest stellar case, in both matrix variants.

The deterministic structural counters expose genuine improvements between the two new strategies. On `stellar_torus/3`, with residual $k = 6$, the count falls from 426,223 to 742. The output-only method completes `stellar_torus/4`, $k = 9$, in 12,682 units, and `nested_interior_torus/3`, $k = 9$, in 397; the ordered strategy reaches its cap on both. A unit of this instrumented work is not a machine instruction or a portable elapsed-time unit. Counts from the older implementation use different instrumentation and are not directly equated.

### 7.4 Paired timings and ablations

The paired benchmark uses all 117 native full-boundary matrices and 256 seeded relabelings described below, for 373 finite inputs. Each is run seven times with a seeded random ordering of the six methods inside each repetition. This yields 15,666 timed calls. Every complete timed answer is independently checked outside the timer; there are 13,475 such replays. Inconclusive results remain in the raw table.

The environment is CPython 3.12.14 on Linux. Geometry construction, parameter reporting, and certificate replay are outside the timed region; input parsing and search are inside it. The two new algorithms use the same one-million work cap and are each run with rank pruning enabled and disabled. The earlier stopping search uses its own 50,000-node cap; these different counters do not represent equal computational budgets. The machine-readable result retains every raw repetition, status, counter, input hash, and code hash. Microsecond-scale timings characterize this run, not a machine-independent performance promise.

| Method | Success | No forest | Inconclusive | Median ($\mu$s) |
|---|---|---|---|---|
| Earlier two-row solver | 67 | 2 | 37 | 42.80 |
| Earlier stopping search | 92 | 14 | 0 | 43.64 |
| Ordered events | 92 | 14 | 0 | 79.24 |
| Output assignments | 92 | 14 | 0 | 82.86 |

The output-only implementation has higher overhead than the existing stopping search on most small instances. On the maintained corpus its median per-input time is about 83 microseconds, compared with 44 for the incumbent. Across the entire bank, it is faster on 16 of the 372 inputs where both methods complete. The improvement in the parameterized worst-case guarantee is therefore not a justification for replacing the incumbent unconditionally.

| Input | $k$ | Ordered (ms) | Output (ms) | Incumbent (ms) |
|---|---|---|---|---|
| Stellar 1 | 0 | 0.068 | 0.080 | 0.058 |
| Stellar 2 | 3 | 0.111 | 0.129 | 0.143 |
| Stellar 3 | 6 | 309.743 | 0.614 | 0.296 |
| Stellar 4 | 9 | capped | 7.737 | 0.441 |
| Stellar 8 | 21 | capped | capped | 2.271 |
| Nested 1 | 1 | 0.135 | 0.139 | 0.165 |
| Nested 2 | 6 | 0.164 | 0.180 | 0.346 |
| Nested 3 | 9 | capped | 0.369 | 0.607 |

On the $k = 6$ stellar case, the ordered median is 309.743 milliseconds and the output-only median is 0.614 milliseconds, about a 505-fold improvement. The incumbent still completes that input in 0.296 milliseconds. On the final nested case the output-only method takes 0.369 milliseconds, versus 0.607 for the incumbent, while the ordered prototype is capped. Both favorable and unfavorable comparisons belong in the integration decision.

The rank ablation is consequential on the harder stellar inputs. Removing rank pruning changes the output-only $k = 6$ median from 0.614 to 4.317 milliseconds, and makes its $k = 9$ stellar run hit the work cap. It has little or negative constant-factor effect on many tiny cases. The largest stellar instance remains a useful counterweight to favorable benchmarks: the incumbent finishes it in 2.271 milliseconds, while the new output-only solver is still capped.

### 7.5 Relabeling one source and evaluating the bounded prefix

The 256 relabelings permute tetrahedra and local corners of the same five-tetrahedron solid-torus zero cut, using seed 20261012471. The complete finite tree oracle finds a successful

tree in every instance. Native BFS top selection succeeds on 238, and the existing native maximum-weight choice succeeds on all 256. Both new exact solvers also succeed on all 256, with independently replayed saved certificates. These are labeled variants of one manifold, not 256 new knot types. Since maximum weight already covers the entire bank, the bank does not demonstrate additional coverage from the new exact search.

The optional dispatcher in Section 6.4 runs the incumbent for at most $|R| + 1$ search nodes before the FPT fallback. It completes all 490 supplied source/full-boundary and relabeled questions, with 468 successes and 22 exact negatives. Every answer is independently checked. In this bank the incumbent always finishes within the prefix; its maximum instrumented count is nine visited search nodes. Explicit separate controls exercise successful and negative fallback certificates and the sparse huge-namespace route. These results establish dispatcher correctness on the bank, not a measured speedup for the wrapper or a universal nine-node bound.

### 7.6 What can be reproduced without the repository

The archive includes all finite input matrices, both runtime strategies, their independent checkers, the pinned earlier finite comparator, saved certificates, the full paired timing table, and a single reproduction entry point. Those finite experiments require only the Python standard library. Figure generation additionally uses Matplotlib; article building uses a standard LaTeX installation.

Reconstructing the native source geometry again requires the pinned maintained runtime and the two named incoming geometry packages. The included scripts accept explicit paths to them. Their file identities and the source input vectors are retained, so this dependency is reviewable and is not hidden behind the previously generated matrices.

## 8 What the local theorem contributes to a quasi-polynomial bound

The useful improvement is a change of parameter. If the cap graph has $s$ non-exterior vertices, straightforward top-tree search can have depth $s$ and a branching factor depending on the matrix size. The present algorithm permits arbitrary $s$ at each fixed number of nonlinear columns. It gives a polynomial-time local operator for $k = 0, 1, 2, \ldots$ held constant, with one uniform algorithm and a computable parameter factor.

Let $N_i$ be the encoded size of the $i$th supplied graph–matrix problem, and let $k_i$ be its residual triple-column parameter. For a sequence of $Q$ such calls, including unsuccessful calls, the local work is bounded by

$$\sum_{i=1}^{Q} 2^{O(k_i \log(k_i+2))} \operatorname{poly}(N_i). \tag{3}$$

The polynomial exponent is independent of $k_i$. If $N_i \le n^a$ and $k_i \le K(n)$ for every call, this is at most

$$Q\, n^{O(1)} 2^{O(K(n) \log(K(n)+2))}.$$

Two parameter regimes make the scale concrete:

| Uniform residual parameter | Local cost for polynomial-size inputs |
|---|---|
| $K(n) = O(\log n / \log\log n)$ | $n^{O(1)}$ |
| $K(n) = O(\log n)$ | $n^{O(\log\log n)}$ |
| $K(n) = O((\log n)^a)$, fixed $a > 0$ | $\exp(O((\log n)^a \log\log n))\, n^{O(1)}$ |

Under the additional uniform deficit hypothesis of Corollary 4.2, bounded $\Delta$ sharpens the $K(n) = O(\log n)$ line to polynomial local time. It also permits $K(n) = O((\log n)^2)$ within $n^{O(\log n)}$ local time.

Small values of $n$ are absorbed into constants; the first line is an asymptotic assertion as $n$ tends to infinity.

**Theorem 8.1** (Conditional transfer to an entire recognition procedure)**.** *Suppose an exact hierarchy procedure on an $n$-crossing input satisfies:*

1. *it makes at most $\exp(O((\log n)^b))$ calls to the present operator, including calls made after resets and on failed branches;*
2. *each call has encoded size at most $n^a$ and residual nonlinear premise budget at most $O((\log n)^c)$, for fixed $a, b, c$;*
3. *all other work, including surface discovery, source authentication, marked geometric transport, and final certification, has a quasi-polynomial total bound.*

*Then substituting the present exact operator preserves a quasi-polynomial total bound. More precisely, its contribution is*

$$\exp\Big(O((\log n)^b + (\log n)^c \log\log n + \log n)\Big).$$

*Proof.* Apply the general premise-budget extension of Theorem 3.2 to every call, and use the uniform size and parameter bounds in (3). Multiplication by the number of calls adds their logarithmic costs. The sum with the stated quasi-polynomial bound for the other operations remains quasi-polynomial. □

This is an accounting theorem with explicit premises. The finite solver proves none of the three hierarchy hypotheses by itself. In particular, a polynomial encoded-time cut does not bound the number of candidate surfaces or the number of restored hierarchy states. A small certificate does not imply an efficient search for it.

The rank-layer reduction provides an additional structural direction. Under native cycle compatibility, if a scenario leaves $p$ mandatory source groups and $q$ residual demands, then it can succeed only if $p \le r \le p + q$. The difference $r - p$ measures the number of additional independent initial-seed witnesses needed to justify nonlinear activation, beyond the mandatory unary source representatives. Controlling that difference, or decomposing the exceptional columns into independently solvable regions, may be more feasible than bounding every raw column count.

The current source experiments deliberately retain large-parameter cases. Stellar subdivisions give examples with increasing numbers of nonlinear columns while the underlying manifold remains a solid torus. This makes clear why the same topological type, or small normal weight alone, cannot justify assuming that every labelled raw instance has small $k$.

# 9 Further research questions and proposed next steps

1. **Obtain single-exponential dependence on nonlinear columns.** The distinguished-block theorem removes factorial event-order enumeration, but unrestricted Bell partitions still give $2^{O(k \log k)}$. Bounded rank deficit gives a single-exponential special case. Can representative families, algebraic matroid methods, or a smaller exact state space reduce this to $2^{O(k)}$? The required theorem must preserve shared predecessor witnesses and external forest independence. Ordinary subset dynamic programming that records only activated outputs can lose this information.

2. **Exploit the native rank layer before enumerating output assignments.** The Stirling-layer theorem constrains the number of witness blocks. Can the graphic rank and unary condensation force some output choices or prerequisite support requirements in polynomial time? A certificate should make each forced choice replayable and avoid assuming that a greedy surviving row order is optimal.

3. **Use an interaction parameter.** Several nonlinear columns may lie in disjoint regions of the unary condensation. Can they be decomposed by separators or matroid sums while retaining their shared forest constraint? The target is a cost exponential in the largest interacting region, instead of the total number of nonlinear columns.

4. **Choose the lower and upper forests together.** The current theorem fixes the lower 0/1 forest. Varying it changes the matrix and its nonlinear parameter. Is there a source-relative operation that reduces $k$ without creating a larger description or losing boundary markings? The relevant goal is a certified joint optimization, not an unverified algebraic simplification of attaching words.

5. **Implement weighted successful-forest optimization.** The weighted theorem optimizes a linear cost over successful trees. A production version could use retained raw passages or a proved upper bound on later transport work as the objective. It should compare against the existing unconstrained maximum-weight tree, and measure cases where the latter leaves a stopping set.

6. **Recognize a certified matroidal activation regime.** If the closure is itself a matroid closure, an additional intersection gives a polynomial algorithm with no bound on $k$. Can native occurrence data supply a compact certificate of that property? Chordal auxiliary graphs give a concrete sufficient class; arbitrary circular Horn systems do not satisfy the exchange axiom automatically.

7. **Transport the complete boundary pattern.** The matrix certificate retains counts and source labels, while a hierarchy also needs cyclic order, orientations, and marked peripheral data. Construct a compiler that turns every selected tree and peel trace into the required marked homeomorphism with a bound on the accumulated encoding. The currently authenticated port charts are a starting point.

8. **Extend the available cancellation moves.** A complete negative forest certificate isolates a genuine limitation of the fixed raw model. Investigate source-authenticated handle slides, disk-bundle absorption, or controlled retriangulations on those residuals. The progress measure should charge the representation growth as well as the reduction in the number of handles.

9. **Develop smaller negative certificates.** Exhaustive scenario/partition coverage is fixed-parameter sized but can be much larger than a positive trace. Can several Rado deficiencies share one succinct implication or rank certificate? A proposed compression must prove that every omitted scenario is covered, rather than merely retain a log of explored branches.

10. **Prove geometric bounds on the residual parameter.** Does a well-chosen admissible hierarchy admit $k = O(\log n)$, or a polylogarithmic total nonlinear premise budget after each free closure? Counterexamples should distinguish poor labelled triangulations from an unavoidable obstruction across all allowed normalizations.

11. **Account for the entire search history.** Combine the local primitive with constructive versions of surface discovery, hierarchy repair, and the quantitative multi-surface steps in the announced algorithm. A complete bound must include failed branches, restarts, certificate replay, and all input bits created along the way.

12. **Formalize the finite core first.** The closure, scenario, separated-demand, and partition lemmas have finite combinatorial statements and modest dependencies. They are suitable first targets for Lean formalization. The geometric interpretation should be formalized separately, so that verification of the finite search is not mistaken for a formal proof of the manifold compiler.

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